

At the Cutting Edge

Growing up, Ally Togliatti liked seeing how things worked. But it wasn't until she took an elective AP Biology course as a high school senior that an interest was ignited in combining the science of biology with the innovation of engineering. As a result, Ally majored in Materials Science Engineering with a focus in Biomaterials at the University of Illinois.

While an undergraduate, Ally actively participated in Engineering Open House, sharing her love of materials science with thousands of potential engineering students from middle and high schools across Illinois. Ally honed her laboratory skills at Northwestern University where she earned a Master's degree in Biomedical Sciences. Her work at Northwestern included development of a novel technique to spin polymer fibers into a structure modeling the human body's native extracellular matrix. This synthetic structure can be used as a scaffolding system that delivers drugs to a wound site while simultaneously enabling regeneration of tissue within the human body.

Now working at Shirley Ryan AbilityLab in Chicago, Ally sees first-hand how cutting-edge research in regenerative medicine impacts patients with traumatic brain injury, spinal muscular atrophy, and other diseases. She is part of a team developing a model that mimics brain cells' reaction to trauma in the hopes of learning how to treat the effects of various forms of brain injury and neurotrauma. In her role as a research assistant, teamwork and communication are essential.

Ally is excited by the work Shirley Ryan is doing and believes it is important to work at a place that encourages and challenges you to think outside the box. She is humbled by the idea that her work creating new therapeutic models could one day help save a life or even prevent a debilitating disease from ever happening.

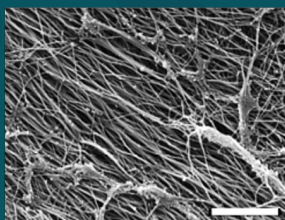


Figure A

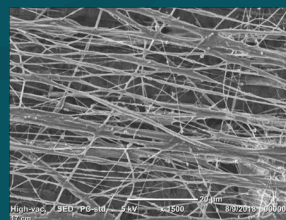


Figure B

A comparison of what the native extracellular matrix looks like in the human body (A) versus the electrospun fiber scaffold (B).

Engineers Changing the World



Ally Togliatti
Research Assistant

"It sounds cliché, but persevere. There were so many times during college and even searching for a job that I wanted to give up. However, working in a place that has the same values as you and encourages you to put your mind to the test is very rewarding. I am still new to Shirley Ryan, but I am tremendously excited for the ample opportunities I have in an environment that's at the cutting edge of research and rehabilitation in my field. So, stick with it! Everything always has a way of working out in the end."

